ARPIT JAIN

DATA SCIENTIST

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SUMMARY

Final-year B.Tech (AI & ML) student specializing in statistical analysis and applied data science: hypothesis testing, feature engineering, and predictive modeling on large, real-world datasets. Delivered a fraud-detection pipeline surfacing significant risk indicators for a business team and an interactive salary-driver dashboard used by non-technical stakeholders. Proficient in SQL, statistical inference, and the Python data-science stack. Open to Data Scientist internship and full-time opportunities focused on analysis, experimentation, and model-driven insight.

TECHNICAL SKILLS

Statistics & Experimentation: Hypothesis Testing, A/B Testing, ANOVA, Chi-Square Tests, Regression, Classification, Clustering, Statistical Inference, Probability

Machine Learning: XGBoost, Ensemble Methods, Collaborative Filtering, SMOTE, Feature Engineering, Dimensionality Reduction (PCA, t-SNE), Cross-Validation, Model Evaluation Metrics

Languages & Libraries: Python, SQL, Pandas, NumPy, Scikit-learn, SciPy, Statsmodels, XGBoost

Data Visualization & BI: Tableau, Power BI, Matplotlib, Seaborn, Plotly, Streamlit

Data Wrangling: SQL Query Optimization, Data Cleaning, Missing-Value & Outlier Treatment, Exploratory Data Analysis (EDA)

Tools: Jupyter Notebook, Git & GitHub, VS Code, Excel

PROJECTS

Financial Fraud Detection & Imbalanced Class Analysis | *Python, Scikit-learn, SMOTE, XGBoost, SQL, Pandas, Statsmodels*

- Achieved 92% recall, 89% precision, and 0.96 ROC-AUC on a 1:99 imbalanced dataset using SMOTE with ensemble classification, reducing false negatives by 15% vs. baseline.
- Extracted and analyzed 500K+ transactions via SQL; identified 7 statistically significant fraud indicators through chi-square testing.

Job Market & Salary EDA | *Python, Pandas, SciPy, Plotly*

- Ran exploratory and inferential analysis on 20K+ StackOverflow/Kaggle salary records; used ANOVA to isolate 3 statistically significant salary drivers ($p < 0.05$): experience level, country, and role type.
- Built an interactive Plotly dashboard enabling non-technical stakeholders to filter and explore findings directly.

CemFormer-X — ML-Driven Manufacturing Optimization | *Python, XGBoost, Pareto Optimization (IEEE manuscript, under review)*

- Built XGBoost surrogate models with <3% prediction error on clinker chemistry, validated across 3 plants, replacing slow lab-assay cycles with real-time inference.
- Designed a Pareto trade-off analysis projecting a 12% cost/emissions reduction; co-authored the statistical methodology in an IEEE-format manuscript.

EXPERIENCE

Machine Learning Research Intern | Suvidha Foundation (CSR Skill Development Programme) *Jan 2026 – Apr 2026*

- Owned surrogate-modeling and statistical-validation work for CemFormer-X, collaborating with an industry mentor and a 5-person research team.

Machine Learning Intern | EliteTech *Jul 2025 – Aug 2025*

- Ran cross-validated model experiments and hyperparameter tuning across candidate algorithms, improving F1-score by 15% on a 100K+ row dataset.
- Engineered and validated predictive features through statistical analysis, cutting data-prep time by 30% via reusable workflows adopted by a 5-person team.

Data Analyst (Python) | Alfido Tech *Jun 2025 – Jul 2025*

- Cleaned and structured raw multi-source data (CSV/JSON) into analysis-ready datasets using Pandas/NumPy across 4 client projects, enabling downstream statistical analysis and reporting.

EDUCATION

B.Tech, Computer Science — AI & ML | Teerthanker Mahaveer University, Moradabad *Aug 2023 – May 2027*

CGPA: 8.5/10 (6th Semester)

Relevant Coursework: Machine Learning, Deep Learning, NLP, DBMS, Statistics, Data Structures

CERTIFICATIONS

- Programming with Generative AI — NPTEL (2024)
- Spoken Tutorial Programme — IIT Bombay (2024) — Git, Python, R — 90%+ Score